

Lecture 05 – Two Pages Report

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Introduction

In this assignment, I extended the COMPAS analysis from class and focused on three different types of machine learning risks: adversarial evasion (PGD attacks), label-flip poisoning, and membership inference.

The main idea was not just to check accuracy, but to see what happens when the model is attacked or stressed in different ways. I used the same models from class (Logistic Regression and Gradient-Boosted Trees) and then tested how they behave under these different scenarios.

Part 1: PGD Evasion Attack

For the first part, I ran a PGD-style attack using epsilon values of 0.25, 0.5, 1.0, and 2.0. The attack is based on the direction of the Logistic Regression coefficients, and I also applied the same adversarial data to the Gradient-Boosted Tree to see if the attack transfers.

What I saw is that as epsilon increases, the false positive rate goes up a lot for both racial groups. At higher values, the model basically starts predicting almost everyone as high risk. So even if the model is technically still running, it is not really useful anymore.

In terms of fairness, the AIR does not go below 0.80 in my results. Instead, it actually moves closer to 1.0. At first this might look like things are becoming “more fair,” but that’s not really true. It’s happening because both groups are being misclassified at very high rates.

One interesting thing is that the Gradient-Boosted Tree is also affected even though the attack was designed for Logistic Regression. That suggests the attack can transfer between models, which makes it more realistic in practice.

Part 2: Poisoning Attack

In the second part, I simulated a label-flip poisoning attack. I tested two cases: targeting African-American defendants and targeting Caucasian defendants. The idea is to flip some of the labels during training so the model learns the wrong patterns.

What stood out here is that AUC does not change that much, even at higher poison rates. For example, when targeting African-American defendants at around 30%, the AUC stays close to the original value. But at the same time, the AIR increases a lot (around 2.8 in my results), which shows a strong fairness issue.

This is what I understood as a “stealth zone.” The model still looks fine if you only check AUC, but fairness is getting worse. That’s a problem because in a real system, this could easily go unnoticed.

I also checked PSI to see if feature drift would detect the attack. It didn't. PSI stayed at zero because the features are not changing, only the labels are. So a normal drift monitor would completely miss this kind of attack.

Part 3: Membership Inference

In the last part, I tested a membership inference attack based on prediction confidence. The idea is that models sometimes give higher confidence to data they were trained on.

In my results, both models have MI AUC values close to 0.50, which basically means the attack is not very effective here. So in this case, there is not much evidence of strong privacy leakage.

That said, the Gradient-Boosted Tree does show a slightly higher MI AUC and a larger generalization gap compared to Logistic Regression. So it might still be a bit more vulnerable.

I also tested different values of the regularization parameter C for Logistic Regression. Stronger regularization slightly reduces the confidence gap, but the effect is not very big in this dataset.

Conclusion

Overall, this assignment showed that models can look fine under normal evaluation but behave very differently under attacks.

The PGD attack showed that the model can become unreliable very quickly. The poisoning attack showed that fairness can get worse even when AUC stays stable, which is probably the most concerning result. The membership inference attack did not show strong risk here, but it still depends on the model and level of overfitting.

I think the main takeaway is that accuracy alone is not enough. We also need to check fairness, robustness, and privacy, especially in applications like COMPAS where decisions have real consequences.

I completed this assignment myself and used ChatGPT only to help correct wording and minor errors.

